Integration pf sin(5x) dx and integration of sin(5x)•5

Let's solve both step by step in a simple way.

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1. Integration of sin(5x)

We use the standard rule:

$$\int \sin(ax)\,dx = -\tfrac{1}{a}\cos(ax) + C$$

Here, $a = 5$

So,

$$\int \sin(5x)\,dx = -\frac{1}{5}\cos(5x) + C$$

👉 Final Answer:

$$-\frac{1}{5}\cos(5x) + C$$

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2. Integration of sin(5x) · 5

Now the integral is:

$$\int 5\sin(5x)\,dx$$

Take 5 outside:

$$= 5\int \sin(5x)\,dx$$

Now use the previous result:

$$= 5\left(-\frac{1}{5}\cos(5x)\right)$$

$$= -\cos(5x) + C$$

👉 Final Answer:

$$-\cos(5x) + C$$

🔑 Key Idea (Very Important)

- When inside function has a number (5x) → divide by that number
- When same number is multiplied outside → it cancels

⚡ Quick Shortcut

$$\int 5\sin(5x)\,dx = -\cos(5x) + C$$

Because 5 (outside) cancels with 1/5 (from integration).

If you want, I can show you how this comes from chain rule (reverse differentiation)—that's the real concept behind it.

Ok when we integrate sin(5x) why we use thus step multiply and divide by 5

Good question — this is exactly where most students get confused. Let's make it very clear.

🔑 The real reason (not just a trick)

When you integrate $\sin(5x)$, you're actually doing reverse of differentiation (chain rule).

We know:

$$\frac{d}{dx}[\cos(5x)] = -5\sin(5x)$$

👉 Notice carefully:

Derivative of $\cos(5x)$ gives 5·sin(5x), not just $\sin(5x)$

⚠️ The problem

We only have:

$$\sin(5x)$$

But derivative gives:

$$5\sin(5x)$$

So we are missing a 5.

💡 So what do we do?

We "adjust" the expression:

$$\sin(5x) = \frac{1}{5} \cdot (5\sin(5x))$$

Now it matches the derivative form!

🧠 Now integrate

$$\int \sin(5x)\,dx = \frac{1}{5}\int 5\sin(5x)\,dx$$

Now this is easy because:

$$\int 5\sin(5x)\,dx = -\cos(5x)$$

So:

$$= \frac{1}{5}(-\cos(5x))$$

$$= -\frac{1}{5}\cos(5x) + C$$

🎯 Final understanding (VERY IMPORTANT)

We are not randomly multiplying/dividing by 5.

We do it because:

👉 The derivative of inside (5x) is 5
👉 To match the chain rule, we must include that 5
👉 So we adjust using multiply & divide

🔥 Shortcut Rule

Whenever you see:

$$\int \sin(ax)\,dx$$

Just remember:

$$= -\frac{1}{a}\cos(ax) + C$$

💬 Simple one-line intuition

You divide by 5 because:

👉 "The inside (5x) changes 5 times faster, so we slow it down by dividing by 5."

If you want, I can ↪ show this using substitution method (u = 5x) — that will make it even clearer.

Still not understand why we use multiply and divide by 5 technique

Alright — forget the "multiply & divide" trick for a moment. That's what's confusing you. Let's understand it from the root idea in the simplest possible way.

🔁 Think in reverse (Derivative → Integral)

We know this:

$$\frac{d}{dx}[\cos(5x)] = -5\sin(5x)$$

👉 This is a fact from differentiation

❓ Now ask yourself

We want:

$$\int \sin(5x)\,dx = ?$$

So we ask:

👉 "Which function gives sin(5x) when differentiated?"

⚠️ Try directly

If we try:

$$-\cos(5x)$$

Differentiate it:

$$\frac{d}{dx}[-\cos(5x)] = 5\sin(5x)$$

❌ This gives 5 sin(5x)
But we only want sin(5x)

💡 So what's wrong?

We are getting extra 5

So we must remove that extra 5

✅ How to remove it?

Simple logic:

If something becomes 5 times bigger, we divide by 5.

So try:

$$-\frac{1}{5}\cos(5x)$$

🔍 Check it (VERY IMPORTANT STEP)

Differentiate it:

$$\frac{d}{dx}\left[-\frac{1}{5}\cos(5x)\right] = -\frac{1}{5}\cdot(-5\sin(5x))$$

$$= \sin(5x)$$

✅ Now it matches perfectly!

🎯 Final Answer

$$\int \sin(5x)\,dx = -\frac{1}{5}\cos(5x) + C$$

🔑 Now the REAL understanding

👉 We are NOT doing multiply & divide randomly

👉 We are just fixing the extra 5 that appears from chain rule

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💬 One-line intuition

"Derivative of cos(5x) gives extra 5, so in integration we cancel it by dividing by 5."

🔁 Why people say "multiply & divide by 5"

It's just a shortcut way to:

👉 force the expression into a known derivative form

But the real reason is what you just saw:
✔ matching derivative
✔ removing extra factor

If you want, ↪ I can show this using u = 5x substitution, which makes this 100% crystal clear without any tricks.

Ok means we say that if sin(f(x)) •f'(x) and we integrate only sin(f(x))

Yes — you're very close, just one small adjustment is needed to make it 100% correct.

✅ The correct idea

If you have:

$$\sin(f(x)) \cdot f'(x)$$

Then integration is direct:

$$\int \sin(f(x)) \cdot f'(x)\,dx = -\cos(f(x)) + C$$

👉 Because this exactly matches the chain rule (perfect form)

❗ But if you only have:

$$\sin(f(x))$$

Then it is incomplete — you are missing $f'(x)$

🔧 So what do we do?

We adjust it so it matches the correct form.

📌 Example: sin(5x)

Here:

- $f(x) = 5x$
- $f'(x) = 5$

Case 1: Complete form

$$\int \sin(5x) \cdot 5 \, dx$$

This is perfect:

$$= -\cos(5x) + C$$

Case 2: Missing derivative

$$\int \sin(5x) \, dx$$

We are missing 5

So we fix it:

$$= \frac{1}{5} \int 5 \sin(5x) \, dx$$

$$= -\frac{1}{5} \cos(5x) + C$$

🔑 Final rule (this is what you should remember)

- If you see:

$$\sin(f(x)) \cdot f'(x)$$

  👉 Direct integration

- If you see:

$$\sin(f(x))$$

  👉 Divide by $f'(x)$

💬 Simple sentence

👉 "If derivative of inside is missing, divide by it."

⚡ Super shortcut

$$\int \sin(f(x))\,dx = -\frac{1}{f'(x)}\cos(f(x)) + C$$

(works when $f'(x)$ is a constant like 5, 3, etc.)

If you want, I can give you 3–4 practice questions so this becomes automatic in your mind.

E power y and sin e power y multiply and integration it

You're asking about something like:

$$\int e^y \cdot \sin(e^y)\,dy$$

Let's break it clearly and simply.

🔍 Look at the pattern

We compare with the rule:

$$\sin(f(y)) \cdot f'(y)$$

Here:

- $f(y) = e^y$
- $f'(y) = e^y$

👉 And we already have e^y multiplied outside!

So this is a perfect match ✅

🚀 Apply the rule directly

$$\int \sin(f(y)) \cdot f'(y)\, dy = -\cos(f(y)) + C$$

So:

$$\int e^y \sin(e^y)\, dy = -\cos(e^y) + C$$

🎯 Final Answer

$$-\cos(e^y) + C$$

💡 Why this works (simple intuition)

- Inside function: e^y
- Its derivative: e^y
- That same derivative is already present outside

👉 So no need to divide or multiply anything

🔁 Quick check (always good habit)

Differentiate:

$$\frac{d}{dy}[-\cos(e^y)] = \sin(e^y) \cdot e^y$$

✅ Matches perfectly!

🔑 Key takeaway

- If you see:

  $$\sin(\text{something}) \times \text{derivative of that something}$$

  👉 Direct integration
- No adjustment needed ✔️

If you want, I can give you slightly tricky versions where the derivative is missing — that's where students usually get stuck.

Should we use ilate rule

Short answer: No — don't use ILATE here.

❌ Why ILATE is NOT suitable

The ILATE rule (for integration by parts) is used when you have something like:

- $x \cdot e^x$
- $x \cdot \ln x$
- $x \cdot \sin x$

👉 where you cannot directly match a derivative pattern

✅ Your case

$$\int e^y \sin(e^y)\, dy$$

Look carefully:

- Inside function = e^y
- Derivative of e^y = e^y

👉 And that same e^y is already present outside!

🔑 So what method?

Use substitution (u-method), not ILATE.

Let:

$$u = e^y$$

Then:

$$du = e^y\, dy$$

🔄 Replace in integral

$$\int e^y \sin(e^y)\, dy = \int \sin(u)\, du$$

$$= -\cos(u) + C$$

Now put back $u = e^y$:

$$= -\cos(e^y) + C$$

🎯 Final Answer

$$-\cos(e^y) + C$$

🧠 Simple decision rule

- If you see:
  function × derivative of inside → use substitution ✅
- If no pattern fits → then think about ILATE / integration by parts

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💬 One-line intuition

👉 "This is a chain rule pattern, not a product rule problem."

If you want, I can ↪ show you a similar example where ILATE is actually required, so you clearly see the difference.